

High-Performance GEMV on AMD Zen 3 via Architecture-Specific AVX2/FMA Optimization

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Abstract. The General Matrix-Vector Multiply (GEMV) is a fundamental Level-2 BLAS operation appearing throughout scientific computing and numerical linear algebra. GEMV is inherently memory-bound, making effective cache use the dominant optimization factor.

We present a hand-optimized GEMV kernel for AMD Zen 3, implemented using AVX2 and FMA intrinsics. Our single-threaded kernel (ZenGEMV) processes 4 rows simultaneously with dual accumulators per row, creating 8 independent FMA chains that fully saturate the Zen 3 pipeline. Benchmarked against OpenBLAS 0.3.31 and AOCL-BLAS 5.2.0 on an AMD Ryzen 5 5600, ZenGEMV achieves up to $1.40\times$ speedup over OpenBLAS and $1.19\times$ over AOCL-BLAS. Our parallel variant (ZenGEMV-P) achieves up to $3.27\times$ speedup over OpenBLAS on small matrices.

We analyse why dispatch overhead, suboptimal accumulator strategies, and generic design choices in general-purpose BLAS libraries leave performance gains available in architecture-specific workloads.

Keywords: GEMV · BLAS · AVX2 · FMA · AMD Zen 3 · SIMD · OpenMP

1 Introduction

The Basic Linear Algebra Subprograms (BLAS)[1] are a widely adopted set of standardized interfaces for vector (Level-1), matrix-vector (Level-2), and matrix-matrix (Level-3) operations. Among these, the Level-2 BLAS operation GEMV (General Matrix-Vector Multiplication) computes $\mathbf{y} \leftarrow \alpha \mathbf{A}\mathbf{x} + \beta \mathbf{y}$. In computational science, GEMV serves as a core subprogram in a broad range of algorithms, including:

1. The Arnoldi procedure[2], an iterative method for approximating eigenvalues and eigenvectors of large matrices.
2. Token generation in Large Language Models (LLMs), where each decoding step reduces to a sequence of GEMV operations.[3]
3. Iterative linear solvers such as the Generalized Minimal Residual method (GMRES)[4].

GEMV is fundamentally memory-bound. For an $M \times N$ matrix, the operation performs $2MN$ floating-point operations while transferring $O(MN)$ bytes of matrix data, yielding an operational intensity of approximately 0.25 FLOP/byte—well below the ridge point of modern processors (we derive this value formally in Section 2). This memory-bound nature means that the primary optimization target is not compute throughput but rather effective use of the memory hierarchy and minimization of pipeline stalls.

Production BLAS libraries such as OpenBLAS[5] and AOCL-BLAS[6] are designed for generality: they handle arbitrary strides, all combinations of transpose and conjugate operations, and dynamically select microkernels based on detected hardware. This generality introduces a roughly constant dispatch overhead per call. While this overhead is amortized away for large matrices in both GEMM and GEMV, it becomes significant for tiny and small matrices where the compute-to-overhead ratio is low—a regime in which GEMV is frequently invoked, for example during per-token decoding in LLM inference.

This paper investigates whether a hand-optimized kernel, tailored to a specific microarchitecture and constrained use case, can outperform these general-purpose libraries. Our contributions are:

1. An optimized single-threaded GEMV kernel (ZenGEMV) using AVX2 (Advanced Vector Extensions 2)[7] and FMA (Fused Multiply-Add)[8] intrinsics with dual accumulators that fully saturate the Zen 3[9] FMA pipeline.
2. A parallel variant (ZenGEMV-P) with row-parallel decomposition using manual thread work division aligned to SIMD boundaries.
3. A detailed source-level analysis of OpenBLAS and AOCL-BLAS GEMV overhead, explaining where and why performance is lost.

2 Background and Related Work

The GEMV operation computes:

$$\mathbf{y} \leftarrow \alpha A\mathbf{x} + \beta \mathbf{y} \quad (1)$$

where $A \in \mathbb{R}^{M \times N}$ is a matrix with M rows and N columns, $\mathbf{x} \in \mathbb{R}^N$ is the input vector, $\mathbf{y} \in \mathbb{R}^M$ is the output vector, and α, β are scalars. The dimensions M (rows) and N (columns) determine both the computational cost and the memory footprint of the operation.

The operation requires $2MN$ floating-point operations—one multiply and one add per element of A . The total data movement is $(MN + N + 2M) \times 8$ bytes for double-precision (the matrix A , input vector $\mathbf{x}$, and the output vector $\mathbf{y}$ read and written), giving an operational intensity[10] of:

$$\text{OI} = \frac{2MN}{8(MN + N + 2M)} \approx \frac{1}{4} \text{ FLOPs/byte} \quad (2)$$

for large M and N , where the vector terms become negligible relative to the matrix. This makes GEMV memory-bound on all modern architectures.

Our target platform is the AMD Ryzen 5 5600, based on the Zen 3 microarchitecture. The relevant parameters are sourced from the AMD Software Optimization Guide[11] and Agner Fog’s instruction tables[12] and microarchitecture guide[13]:

- **Cache hierarchy:** 32 KiB L1d (8-way, latency 4 clocks) per core, 512 KiB L2 (8-way, latency 14 clocks) per core, 32 MiB shared L3 (16-way, latency 47 clocks).
- **Execution units:** Two 256-bit FMA units capable of executing `vfmadd` instructions.
- **FMA pipeline:** Each `vfmadd231pd` instruction has a 4-cycle latency but the two FMA units can accept a new instruction every cycle (2-per-cycle throughput). Because the pipeline is 4 stages deep, an FMA result is not available for 4 cycles after issue. To avoid stalls, the code must maintain independent operations that can fill the pipeline while earlier results are computed. With 2 units each having 4-cycle latency, the minimum number of independent FMA operations needed for full utilization is $4 \times 2 = 8$.
- **Peak throughput:** $2 \text{ FMA} \times 4 \text{ doubles} \times 2 \text{ ops} \times 4.4 \text{ GHz} = 70.4 \text{ GFLOPS/core}$ at maximum boost clock. This value is the theoretical peak.
- **Memory bandwidth:** The Ryzen 5 5600 uses a DDR4 memory controller. Our test system is equipped with DDR4 3200 MT/s CL16 modules, providing a theoretical peak of 51.2 GB/s in dual-channel mode. This value was calculated using an online tool.[14]

The Roofline model[10] provides an upper bound on attainable performance as a function of a kernel’s operational intensity (OI, the ratio of floating-point operations to bytes transferred) and the system’s memory bandwidth (BW, in GB/s):

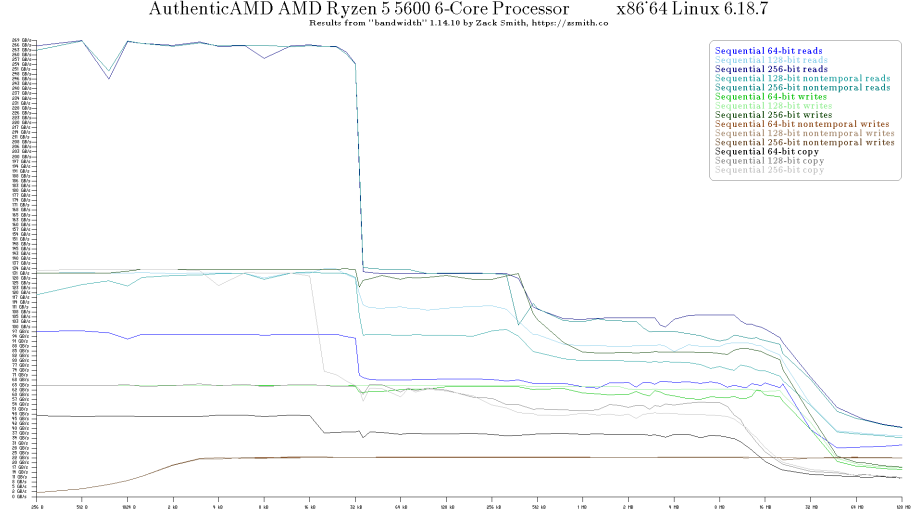
$$P_{\max} = \min(P_{\text{peak}}, \text{BW} \times \text{OI}) \quad (3)$$

For GEMV with $\text{OI} \approx 0.25 \text{ FLOP/byte}$, the theoretical maximum depends on which level of the memory hierarchy serves the data. The practical bandwidth at each cache level is derived from the sustained transfer width per cycle at the sustained core frequency of 4.4 GHz:

- **L1d:** The L1d cache supports up to two 256-bit (32-byte) loads and one 128-bit (16-byte) load simultaneously which combined with the 4.4 GHz boost clock gives $2 \times 32 \times 4.4 + 1 \times 16 \times 4.4 = 352 \text{ GB/s}$ theoretical throughput; practical throughput is likely less than $\sim 281.6 \text{ GB/s}$ due to at least one port being usually dedicated to store operations.[11]
- **L2:** The L2 to L1 data path delivers 32 bytes/cycle to L1, giving $32 \times 4.4 = 140.8 \text{ GB/s}$.[11]
- **L3:** The per-core L3 read bandwidth was measured empirically using the *bandwidth* tool by Zack Smith [15] on sequential 256-bit reads. Buffer sizes in the L3-resident range (1 MB–16 MB) yielded $\sim 106 \text{ GB/s}$ sustained read bandwidth. The raw measurement output is shown in Figure 1.
- **DRAM:** DDR4-3200 dual-channel provides $3200 \times 8 \times 2 = 51.2 \text{ GB/s}$ theoretical; practical $\sim 42 \text{ GB/s}$ after controller overhead.

Table 1. Bandwidth ceilings and GEMV performance limits per memory level (OI = 0.25 FLOP/byte).

Memory Level	Practical BW	GEMV Ceiling	Size
L1d	267 GB/s	66.8 GFLOPS	32 KB
L2	132 GB/s	33.0 GFLOPS	512 KB
L3	106 GB/s	26.5 GFLOPS	32 MB
DRAM	42 GB/s	10.5 GFLOPS	—

**Fig. 1.** Sequential 256-bit read bandwidth measured with the *bandwidth* tool [15] on the AMD Ryzen 5 5600. The four plateaus correspond to L1d (~ 267 GB/s), L2 (~ 132 GB/s), L3 (~ 106 GB/s), and DRAM (~ 42 GB/s).

The GEMV ceiling at each level is then computed from Equation (3) as $\text{BW} \times \text{OI}$ (e.g., $267 \times 0.25 = 66.75$ GFLOPS for L1d). Table 1 summarizes these values.

When the working set fits in cache, the effective bandwidth increases dramatically, enabling much higher throughput for small to medium matrices than the DRAM-bound prediction would suggest. Figure 2 shows the roofline diagram with our measured ZenGEMV performance for each benchmark preset, demonstrating how cache residency determines the reachable performance level.

2.1 Related Work

OpenBLAS[5] descends from GotoBLAS[16] and provides architecture-specific assembly kernels. Its GEMV implementation uses column blocking with $\text{NBMAX}=2048$. All ZEN generations share the microkernel `dgemv_t_microk_haswell-4.c`, which processes 4 rows simultaneously with a single accumulator per row.

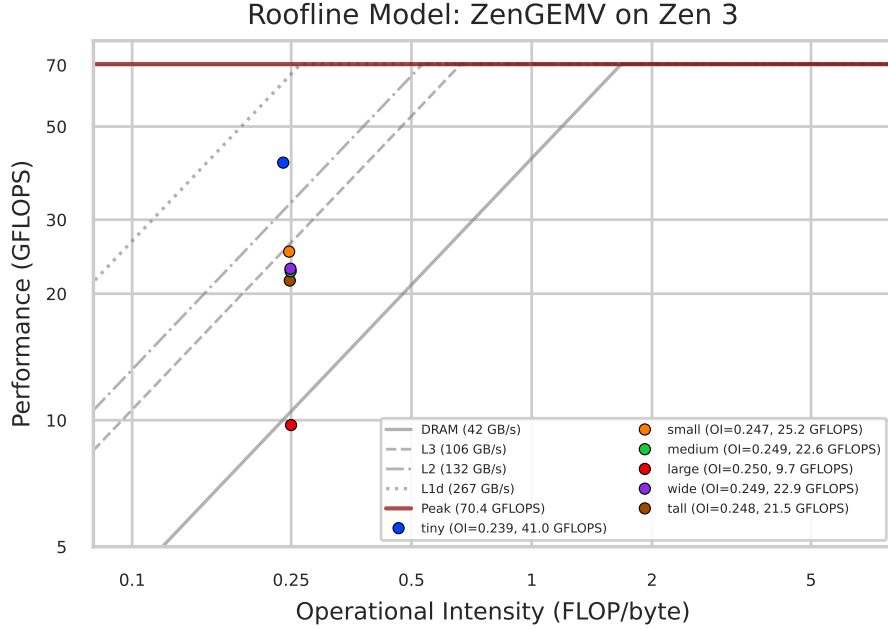


Fig. 2. Roofline model for Zen 3 showing bandwidth ceilings from DRAM through L1d cache (diagonal lines) and the compute peak (horizontal line). Colored points show measured ZenGEMV kernel performance at each preset’s operational intensity. Matrices whose working set fits in L1 or L2 cache achieve substantially higher throughput than the DRAM-bound prediction.

AOCL-BLAS is AMD’s optimized fork of the BLIS[17] framework, decomposes GEMV operations into fused Level-1 microkernels with a configurable fuse factor (detailed in Section 4), and includes a Zen 3 tuned configuration in version 5.2.0.

The ATLAS project[18] pioneered empirical auto-tuning for BLAS, searching blocking parameters and code transformations to maximise cache efficiency. Hassan[19] demonstrates that hand-written AVX SGEMV kernels can outperform Intel MKL on multicore x86 processors; Fog[12,13] documents detailed instruction latencies and throughput for x86 microarchitectures, which informed our kernel design.

3 Implementation

The ZenGEMV kernel processes 4 rows of the matrix simultaneously in the inner loop. This design is motivated by two key constraints of the Zen 3 FMA pipeline:

1. **FMA latency hiding:** Each `vmadd231pd` instruction has 4-cycle latency[12]. With 2 FMA units executing per cycle, we need $4 \times 2 = 8$ independent FMA operations to keep both units fully occupied.
2. **reusing vector x:** Loading the `x` vector once and applying it to 4 rows reduces memory traffic.

For each of the 4 rows, we maintain two independent accumulator registers (`sum_a` and `sum_b`).

The main loop processes 16 elements per iteration (4 AVX2 loads of 4 doubles each), alternating between accumulators `a` and `b`:

```

1 for (; j + 15 < cols; j += 16) {
2     __m256d x0 = _mm256_load_pd(&x[j]);
3     __m256d x1 = _mm256_load_pd(&x[j + 4]);
4     __m256d x2 = _mm256_load_pd(&x[j + 8]);
5     __m256d x3 = _mm256_load_pd(&x[j + 12]);
6
7     sum0a = _mm256_fmadd_pd(
8         _mm256_load_pd(&a0[j]), x0, sum0a);
9     sum0b = _mm256_fmadd_pd(
10        _mm256_load_pd(&a0[j + 4]), x1, sum0b);
11    sum0a = _mm256_fmadd_pd(
12        _mm256_load_pd(&a0[j + 8]), x2, sum0a);
13    sum0b = _mm256_fmadd_pd(
14        _mm256_load_pd(&a0[j + 12]), x3, sum0b);
15    // ... rows 1-3 follow the same pattern
16 }
```

Listing 1.1. Inner loop of ZenGEMV (abbreviated, showing 1 of 4 rows).

After the main loop, the dual accumulators are combined, and cleanup loops handle remaining elements in groups of 8, 4, and finally scalar elements. A horizontal sum reduction (`hsum_pd`) converts each 256-bit accumulator to a scalar result using `vextractf128` and `vhaddpd`[7].

Compiler hints including GCC's `always_inline` and `hot` attributes are used to ensure the kernel is inlined at the call site and marked as performance-critical for the compiler's optimization heuristics. The `__restrict` qualifier on all pointer parameters enables the compiler to assume no aliasing.

The compilation flags `-O3 -march=znver3 -mfma -mavx2 -ffast-math -flto`[20] enable architecture-specific code generation and link-time optimization.

The ZenGEMV-P variant adds multi-threaded execution using OpenMP[21] with a row-parallel decomposition strategy. Rather than using `#pragma omp parallel for`, which incurs per-iteration scheduling overhead, we use a single `#pragma omp parallel` region with manual work division:

```

1 #pragma omp parallel
2 {
3     int nt = omp_get_num_threads();
4     int tid = omp_get_thread_num();
5     int total_chunks = (rows + 3) / 4;
6     int chunks_per = total_chunks / nt;
7     int extra = total_chunks % nt;
8     int start = tid*chunks_per + min(tid, extra);
9     int end = start + chunks_per + (tid < extra);
10    process_row_range(start*4, min(end*4, rows),
11                      cols, alpha, A, x, y);
12 }
```

Listing 1.2. Thread work division in ZenGEMV-P.

Key design decisions:

- **4-row alignment:** Thread boundaries are aligned to 4-row groups, ensuring each thread uses the optimized 4-row kernel without partial processing at boundaries.
- **No false sharing:** Each thread writes to a disjoint region of $\mathbf{y}$, eliminating cache-line contention.
- **Adaptive thresholds:** Parallelization is only applied when the matrix has at least 128 rows and between 64K and 8M elements. Below the lower threshold, thread creation overhead dominates. Above the upper threshold, DRAM bandwidth saturation makes additional threads counterproductive—measurements confirm that serial ZenGEMV outperforms the parallel variant in this regime, consistent with DRAM bandwidth saturation predicted by the roofline model (Figure 2).

4 Analysis of Library Overhead

OpenBLAS GEMV processing involves several layers of overhead before reaching the kernel:

1. **CBLAS dispatch:** `cblas_dgemv` validates the memory layout (`CblasRowMajor/CblasColMajor`) and transpose flag (`NoTrans/Trans/ConjTrans`), swapping M/N and the transpose sense before forwarding to the Fortran-style `dgemv` entry point.
2. **Column blocking:** The implementation splits columns into blocks of `NBMAX=2048`, performing a horizontal reduction after each block. For matrices with $N > 2048$, this introduces additional reduction overhead and breaks the accumulation across blocks.
3. **Accumulator strategy:** OpenBLAS compiles all ZEN generations (ZEN through ZEN4) using the same shared microkernel `dgemv_t_microk_haswell-4.c`, as determined by the preprocessor block in `kernel/x86_64/dgemv_t_4.c`:

```
#if defined(HASWELL) || defined(ZEN) || ...
#include "dgemv_t_microk_haswell-4.c"
#endif
```

This kernel processes 4 rows simultaneously with a single accumulator per row, providing only 4 independent FMA chains. On Zen 3, 8 chains are needed for full utilization—this kernel leaves half the pipeline capacity unused.

AOCL-BLAS uses an object-based design that provides clean abstractions at a performance cost:

1. **Object dispatch:** `bli_dgemv` constructs `obj_t` structures (AOCL-BLAS’s internal matrix/vector descriptors carrying dimensions, strides, and data pointers) and queries the hardware context (`cntx_t`—a structure storing function pointers to architecture-specific microkernels along with cache and register blocking parameters for the target platform).
2. **Kernel abstraction:** AOCL-BLAS decomposes the GEMV operation into fused Level-1 primitives rather than implementing it monolithically. For the non-transpose case (`bli_gemv_unf_var2`), it uses **AXPYF**—a fused `axpy` that performs f scaled vector additions in a single pass, with fuse factor `b_fuse=BLIS_AF = 5` on Zen 3. For the transpose case (`bli_gemv_unf_var1`), it uses **DOTXF**—a fused dot product that simultaneously computes the dot product of $\mathbf{x}$ against f matrix rows, with `b_fuse=BLIS_DF = 8` on Zen 3. While this modularity enables code reuse across different BLAS operations, it prevents optimizations that span the entirety of the GEMV operation.
3. **Per-call initialization:** Every entry through the BLAS compatibility layer invokes `bli_init_auto()`—a library initialization guard—followed by parameter mapping (`bli_param_map_netlib_to_blis_trans`), dimension extraction, and stride setup before any computation begins. The kernel itself is invoked through a function pointer retrieved from `cntx` at runtime, adding an indirect call on every GEMV invocation.

Despite these overheads, general-purpose libraries have advantages in certain scenarios:

- **AOCL-BLAS on medium (1T):** AOCL-BLAS achieves 23.46 GFLOPS vs. our 22.62 GFLOPS on 1024×1024 . The AOCL-BLAS Zen 3 kernel configuration is well-tuned for L3-resident data, and its grouping strategies may provide better cache utilization at this specific size.
- **Multi-threaded medium/tall presets:** OpenBLAS and AOCL-BLAS achieve competitive or slightly better multi-threaded performance on medium and tall presets, where the larger amount of work per call amortizes their thread-dispatching overhead.
- **Generality:** Libraries handle arbitrary strides, non-contiguous memory layouts, complex alpha/beta combinations, and all transpose variants—features our kernel does not support.

5 Experimental Evaluation

All experiments were conducted on an AMD Ryzen 5 5600 (6 cores, 12 threads, Zen 3, base clock 3.5 GHz, boost up to 4.47 GHz) running Arch Linux with kernel 6.18.7-zen1-1-zen, with the following configuration:

- **Clock source:** `CLOCK_MONOTONIC_RAW` to avoid NTP adjustments.
- **Measurement:** 5 runs per configuration, reporting median GFLOPS with standard deviation, minimum, and maximum.

Table 2. Benchmark presets with working set sizes and cache residency. Working set computed as $(MN + N + 2M) \times 8$ bytes.

Preset	M	N	Working Set	Cache Level
Tiny	64	64	33.5 KB	L1d
Small	256	256	518.0 KB	L2
Medium	1024	1024	8.02 MB	L3
Large	4096	4096	128.1 MB	DRAM
Wide	256	8192	16.1 MB	L3/DRAM
Tall	8192	256	16.1 MB	L3/DRAM

Table 3. Single-threaded GEMV performance (median GFLOPS, 5 runs). Speedup columns show ZenGEMV divided by the respective library’s throughput.

Preset	ZenGEMV	OpenBLAS	AOCL-BLAS	ZenGEMV / OpenBLAS	ZenGEMV / AOCL-BLAS
Tiny	41.02	29.30	35.09	1.40×	1.17×
Small	25.19	23.28	21.13	1.08×	1.19×
Medium	22.62	22.14	23.46	1.02×	0.96×
Large	9.74	9.01	8.82	1.08×	1.10×
Wide	22.94	21.73	22.96	1.06×	1.00×
Tall	21.50	20.14	21.69	1.07×	0.99×

- **Core pinning:** Single-threaded tests pinned to core 0 via `taskset -c 0`; multi-threaded tests pinned to physical cores 0–5 via `taskset -c 0-5`.
- **Compiler:** GCC 15.2.1 with `-O3 -march=znver3 -mfma -mavx2 -ffast-math -flto`.
- **Libraries:** OpenBLAS 0.3.31 (system package), AOCL-BLAS 5.2.0 (built from source with `-t openmp, zen3` configuration).

Six benchmark presets cover different cache residency scenarios, summarized in Table 2.

Figure 3 shows single-threaded performance across all presets. Table 3 provides detailed numerical results. The speedup columns show the ratio of ZenGEMV performance to each library.

The results reveal a clear pattern tied to cache residency:

Tiny (L1-resident): ZenGEMV achieves 41.02 GFLOPS, a 1.40× speedup over OpenBLAS. At this size, the matrix, vector, and result all fit in L1d cache, making dispatch overhead significant relative to computation time. ZenGEMV’s direct function call with inlined kernel eliminates this overhead entirely.

Small (L2-resident): ZenGEMV achieves 25.19 GFLOPS (1.08× over OpenBLAS, 1.19× over AOCL-BLAS). At this size, dispatch overhead is a smaller fraction of total execution time, bringing results from all implementations closer together.

Medium (L3-resident): AOCL-BLAS achieves the highest throughput at 23.46 vs. 22.62 GFLOPS for ZenGEMV. The AOCL-BLAS Zen 3 configuration

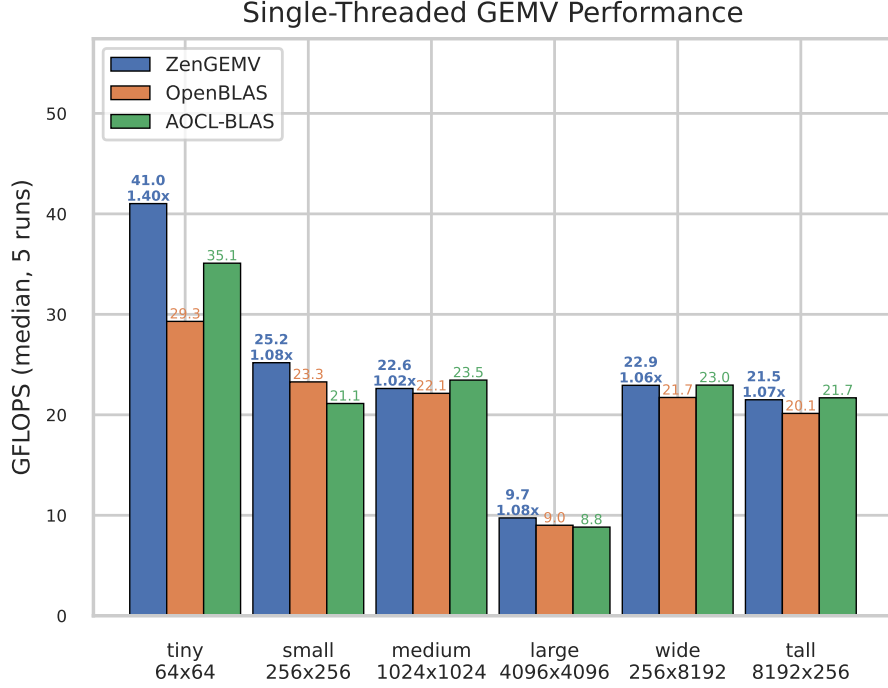


Fig. 3. Single-threaded GEMV performance comparison. Annotations above ZenGEMV bars show throughput (GFLOPS) and speedup relative to OpenBLAS.

appears better suited to this cache level, outperforming both OpenBLAS and ZenGEMV.

Large (DRAM-bound): All implementations converge near 10 GFLOPS, consistent with the DRAM roofline prediction of ~ 10.5 GFLOPS (Table 1). At this point, memory bandwidth is the bottleneck and kernel optimizations yield diminishing returns.

Figure 4 shows multi-threaded performance with 6 physical cores. Table 4 provides numerical details.

Key observations:

Tiny and small presets: OpenBLAS deliberately runs single-threaded for $MN < 115,200 \times T$ (where $T = 4$ is its compile-time multithread threshold, giving a cutoff of 460,800). Both the tiny ($MN = 4,096$) and small ($MN = 65,536$) presets fall well below this limit. The rationale is that for small matrices it is more efficient to issue multiple independent GEMV calls concurrently at the application level than to parallelize within a single call. ZenGEMV-P and AOCL-BLAS both choose to parallelize within the call: on the small preset ZenGEMV-P reaches 77.43 GFLOPS ($3.07\times$ over its single-threaded result) versus AOCL-

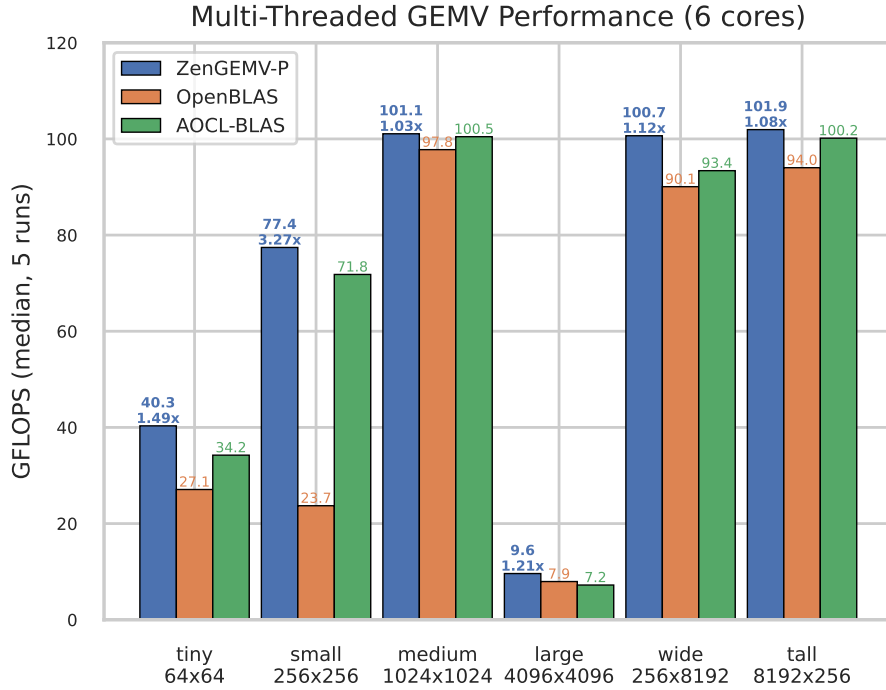


Fig. 4. Multi-threaded GEMV performance (6 physical cores). Annotations above ZenGEMV-P bars show throughput (GFLOPS) and speedup relative to OpenBLAS.

BLAS at 71.81 GFLOPS ($3.40\times$ over its 21.13 GFLOPS), a $1.08\times$ speedup for ZenGEMV-P.

Large preset (serial fallback): ZenGEMV-P falls back to serial execution for matrices exceeding 8M elements, achieving 9.61 GFLOPS. Both OpenBLAS (7.95) and AOCL-BLAS (7.22) attempt parallelization but suffer from memory controller contention, performing worse than the serial case.

Medium/Wide/Tall: These L3-resident sizes show competitive performance across all implementations. ZenGEMV-P leads at medium ($1.03\times$ over OpenBLAS, $1.01\times$ over AOCL-BLAS), wide ($1.12\times$, $1.08\times$), and tall ($1.08\times$, $1.02\times$).

Table 5 compares measured performance against roofline predictions at each cache level.

The large preset achieves 92.8% of the DRAM roofline, confirming that our kernel is effectively bandwidth-limited with minimal computational overhead. The L3-resident presets (medium, wide, tall) achieve 81–87% of their ceiling, indicating efficient cache utilization with some loss to cache miss penalties at boundaries between cache tiers. The L1-resident tiny preset achieves 62.0% of its ceiling, with the gap attributable to horizontal reduction overhead, loop control, and the fact that L1 bandwidth is shared between loads and stores.

Table 4. Multi-threaded GEMV performance (median GFLOPS, 6 cores, 5 runs). Speedup columns show ZenGEMV-P divided by each library.

Preset	ZenGEMV-P	OpenBLAS	AOCL-BLAS	ZenGEMV-P / OpenBLAS	ZenGEMV-P / AOCL-BLAS
Tiny	40.33	27.09	34.23	1.49×	1.18×
Small	77.43	23.71	71.81	3.27×	1.08×
Medium	101.07	97.77	100.46	1.03×	1.01×
Large	9.61	7.95	7.22	1.21×	1.33×
Wide	100.66	90.07	93.39	1.12×	1.08×
Tall	101.93	94.01	100.15	1.08×	1.02×

Table 5. Roofline predictions compared to measured ZenGEMV performance

Preset	OI BW Ceiling (FLOP/B)	Predicted (GB/s) (GFLOPS)	Measured (GFLOPS)	% of Ceiling
Tiny	0.248	267 (L1d)	66.3	41.02
Small	0.249	132 (L2)	32.9	25.19
Medium	0.250	106 (L3)	26.5	22.62
Large	0.250	42 (DRAM)	10.5	9.74
Wide	0.249	106 (L3)	26.4	22.94
Tall	0.250	106 (L3)	26.5	21.50

6 Discussion and Future Work

Our results demonstrate that architecture-specific kernel optimization yields measurable performance improvements over general-purpose BLAS libraries for Level-2 operations where dispatch overhead is non-negligible relative to compute time. Several avenues for further improvement remain:

Custom kernels at small sizes: Production BLAS libraries avoid intra-call parallelism for small matrices, assuming the application will issue multiple calls concurrently. When this assumption does not hold, a hand-optimized kernel is the natural solution: it eliminates dispatch overhead and can exploit all available threads without requiring application-level batching. Our results on the tiny and small presets confirm that dispatch overhead savings alone yield 1.40× and 1.08× speedups over OpenBLAS in the single-threaded case.

Assembly implementation: While intrinsics provide portability, the compiler still makes register allocation and scheduling decisions that may be suboptimal. A hand-written assembly kernel could precisely control instruction ordering and register usage, potentially closing the gap to the roofline on L1-resident data.

Shape-adaptive dispatch: Our kernel uses the same strategy for all matrix shapes. A runtime dispatch based on the aspect ratio (wide vs. tall vs. square) could select between row-parallel and column-parallel decomposition strategies, potentially improving cache utilization for extreme aspect ratios.

It should be noted that the current implementation imposes constraints not present in general-purpose libraries: row-major layout, 32-byte aligned memory, and unit alpha/beta scalars. Extending support to arbitrary strides and memory layouts would require additional code paths that may reduce the observed performance advantage.

7 Conclusion

We have presented ZenGEMV and ZenGEMV-P, optimized GEMV kernels for the AMD Zen 3 microarchitecture that outperform OpenBLAS and AOCL-BLAS across most benchmark configurations. The key insight is that Zen 3’s dual FMA units with 4-cycle latency require 8 independent operations in flight, achieved through our 4-row $\times$ 2-accumulator design. This provides strictly higher pipeline utilization than OpenBLAS’s 4-chain Haswell microkernel.

In single-threaded mode, ZenGEMV reaches up to $1.40\times$ over OpenBLAS and $1.19\times$ over AOCL-BLAS, with the largest advantage on L1-resident data where dispatch overhead dominates. In multi-threaded mode, ZenGEMV-P scales efficiently across all 6 cores and reaches a $3.27\times$ advantage over OpenBLAS on the small preset where OpenBLAS deliberately serializes.

These results indicate that for fixed-platform deployments in high-performance and computational science contexts where GEMV constitutes a performance-critical operation, architecture-specific hand optimization remains a viable and productive strategy, even alongside mature general-purpose libraries. Future work includes a hand-written assembly kernel to close the remaining gap to the roofline ceiling, shape-adaptive dispatch for non-square matrices, and extended support for arbitrary memory layouts and scaling parameters.

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